

Sudharshan Ravi

Zürich, Switzerland

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Summary

- **My Profile:** Computational Researcher at the University of Lausanne (UNIL) | Ph.D. in Chemical and Bioengineering (Focus on Computational Systems Biology) at ETH Zürich | Swiss resident (valid work permit) | Indian citizen | 10+ years in Bioinformatics Research | 7+ years in Metabolic Network Analysis | 5+ years as Lead Researcher
- **My work:** Integrative high-dimensional multi-omics data analysis in understanding the biology of aging and age-related diseases | Novel workflow for the analysis and integration of omics data in genome-scale metabolic models | Inferential analysis on the impact of socio-economic and other exogenous gradients on transcriptomic patterns and health outlook

Experience

Premier Assistant

Lausanne, Switzerland

DEPARTMENT OF FUNDAMENTAL MICROBIOLOGY (DMF), UNIVERSITY OF LAUSANNE (UNIL)

Mar. 2026 – Present

- Developing constraint-based computational models to characterize metabolic interactions and resource exchange networks in microbial communities, advancing mechanistic understanding of their emergent functional dynamics.

Bioinformatics Lead

London, UK

MYOLYTICS

Jan. 2025 – Feb. 2026

- Tailored machine learning models to longitudinal multimodal molecular and biosensor data for pre-symptomatic prediction of muscle injury risk, translating predictive insights into personalized management strategies for professional athletes.

Postdoctoral Research Associate

Zürich, Switzerland

SOCIAL GENOMICS, JACOBS CENTER FOR PRODUCTIVE YOUTH DEVELOPMENT, UZH ZÜRICH

Mar. 2021 – Jan. 2025

- Characterized molecular consequences of socioeconomic inequalities across transcriptomic, immunological and metabolic axes using [multi-omics and network-based approaches](#) on a longitudinal cohort spanning adolescence to adulthood.
- Revealed disruption of upstream immune regulatory networks as a key mechanism connecting early-life socioeconomic disparities to altered metabolic function, accelerated aging and increased disease predisposition.

Graduate Research Assistant / Doctoral Candidate

Zürich, Switzerland

CHEMICAL AND BIOLOGICAL SYSTEM ENGINEERING LABORATORY, ETH ZÜRICH

Jan. 2016 – Sep. 2020

- [Bioinformatic analysis of human aging](#): Extensive analysis of population-level transcriptomic data characterizing the biology of aging and its close relationship to metabolism.
- [ΔFBA](#): Developed a novel pipeline integrating differential omics into genome-scale metabolic models using constraint-based modeling and mathematical optimization.
- [Multi-omics analyses](#): Deciphered the chromatin – gene expression interplay in cell fate commitment of hematopoietic cells. [Illuminated](#) the mechanisms of constrained antitumor immune responses during IDO1 inhibition in ovarian cancer.

Research Assistant

Boston, USA

SENGUPTA'S LAB FOR NANOMEDICINE, HARVARD – MIT HEALTH SCIENCES AND TECHNOLOGY

Jan. 2012 – Aug. 2012

- [Elucidated](#) CD44v6–EGFR scaffold kinase interaction as a mechanism predisposing chemotherapy-tolerant breast cancer cells and a bivalent therapeutic target for overcoming drug resistance.

Education

Ph.D. in Chemical and Bioengineering

Zürich, Switzerland

ETH ZÜRICH | VISITING RESEARCHER AT UNIVERSITY AT BUFFALO, USA (Aug. 2018 – Jul. 2020)

Jan. 2016 – Sep. 2020

- Thesis: [Metabolic network analysis and its application in understanding the biology of aging](#) – Prof. Dr. Rudiyanto Gunawan
- Research Specialization: Bioinformatics | Systems Biology | Aging & Age-Related Diseases | Metabolic Network Analysis | Constraint-Based Modeling | Integrative Multi-omics

M.Sc. in Chemical and Bioengineering

Zürich, Switzerland

ETH ZÜRICH

Sep. 2013 – Sep. 2015

- Thesis: [Reducing Complexity – Efficient Modeling for Biorefinery Concepts](#) – Prof. Dr. Konrad Hungerbühler
- Research Specialization: Surrogate Modeling | Sustainable Engineering | Machine Learning | Bioprocessing Technologies | Process Engineering

B.Tech. in Industrial Biotechnology

Thanjavur, India

SASTRA UNIVERSITY | HARVARD – MIT HEALTH SCIENCES AND TECHNOLOGY

Sep. 2008 – Oct. 2012

- Thesis: [An Induced CD44v6 – EGFR scaffold kinase interaction predisposes a subpopulation of chemotherapy-tolerant breast cancer cells and serves as a bivalent target of resistance](#) – Prof. Dr. Aaron Goldman
- Research Specialization: Molecular Biology | Genetics | Cancer Biology | Drug Resistance | Nanomedicine

Selected Publications

- [1] **Sudharshan Ravi**, Michael J. Shanahan, Brandt Levitt, *et al.* Socioeconomic inequalities in early adulthood disrupt the immune transcriptomic landscape via upstream regulators. **Scientific Reports**, 2024 [↗](#)
- [2] **Sudharshan Ravi** and Rudiyanto Gunawan. Δ FBA – Predicting metabolic flux alterations using genome-scale metabolic models and differential transcriptomics data. **PLOS Computational Biology**, 2021 [↗](#)
- [3] Emelyne Teo, **Sudharshan Ravi**, Diego Barardo, *et al.* Metabolic stress is a primary pathogenic event in transgenic *Caenorhabditis elegans* expressing pan-neuronal human amyloid beta. **eLife**, 2019 [↗](#)

See complete list of publications and conference presentations at [Google Scholar](#).

Proficiencies

COMPUTATIONAL	 Bioconductor Python scikit-learn pandas MATLAB COBRA Toolbox
INFRASTRUCTURE	Bash Git Docker High-Performance Computing
NETWORKS	Cytoscape STRING iGraph Gephi
VISUALIZATION	ggplot2 Adobe Illustrator Inkscape PowerPoint $\text{\LaTeX}$
LABORATORY	Cell culture IHC Western Blotting Flow cytometry Fluorescence microscopy
LANGUAGES	English (Fluent) German (Intermediate) Tamil (Native) Hindi (Beginner)

Academic Merits & Activities

SCHOLARSHIPS

- 2015 Birkigt Scholarship Award, ETH ZÜRICH
- 2012 Undergraduate Research Scholarship, SASTRA UNIVERSITY & HARVARD – MIT HST

MENTORING

- 2016 – 2018 Tutored [Bioprocess Modeling](#), [Process Simulation and Flowsheeting](#) and [Statistical and Numerical Methods](#) courses at ETH ZÜRICH
- 2016 – 2019 [Supervised](#) 2 MS students in research projects and semester theses at ETH ZÜRICH & UNIVERSITY AT BUFFALO